

P2 - CSG - Sheet and Print

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Revision: 2026-09-22. Dates and weights follow the course schedule.

P2a - CSG process and instructions

Introduced: Week 2 - Sep 1 / **Due:** Week 4 - Sep 15 / **Weight:** 7%

Description

Build an editable CSG object and explain its construction as a sequence another person can follow. Use a BIG-style massing diagram: one operation per frame, a consistent viewpoint and a visible change between steps. The same object becomes the basis of the P2b print.

Rules & Constraints

- One 17 x 11 inch sheet laid out as a BIG-style process diagram (in the manner of Bjarke Ingels Group's step-by-step massing diagrams): eight or nine numbered frames, one operation per frame, read in sequence.
- Use at least three primitives; include translation, rotation and scale, plus union, difference and intersection. Label the kept and removed solids in each subtraction.
- Draw the final solid at building scale with a 1.75 m figure, vegetation, ground and shadow.
- Add clear instructions and matching pseudocode beside the diagrams, one step per operation, with the exposed parameters.
- Refer to Di Mari and Yoo's Operative Design and Di Mari's Conditional Design; name the spatial operation your sequence explores.
- Submit an editable graph with exposed inputs and all source operations intact.

Parameters

- Expose primitive dimensions, translation values, rotation angle/axis/center and scale factor. Name inputs by their geometric role and state units.
- Define the origin and axes, where each primitive starts, and the order of Boolean operands. In a difference, identify the solid kept and the cutter removed.
- Keep the source operations editable. Preserve native node names and use Frames/groups to explain the sequence. Use Blender Geometry Nodes or an already approved Rhino/Grasshopper route.

Required Studies

- **Eight or nine numbered frames.** Begin with at least three primitive solids. Include translation, rotation, scale, union, intersection and difference. Mark incoming geometry and removed material so the change can be read without guessing. Each frame records one operation or operation type, with its relevant values.
- **Matching instructions and pseudocode.** Give intermediate solids names, use those names consistently and place the text beside its frame. State the units and coordinate conventions once, then specify enough dimensions and transforms to reconstruct the result.
- **Final solid at building scale.** Show the final geometry with a 1.75 m figure, vegetation, ground and shadow. Separate presentation context from the CSG object that will be fabricated. State the spatial operation explored, with a reference to Operative Design or Conditional Design.

Workflow

1. Choose a spatial operation from the readings, sketch a sequence and build it from primitives using an editable graph.
2. Check the Boolean order and change an exposed input. Verify that the downstream result recomputes without rebuilding the graph.
3. Write the procedure from the working model. Ask a partner to identify any undefined size, position or operand; revise those instructions.
4. Export the operation views from one camera and compose the single sheet using the Graphic Standards Manual. Match every frame to the current graph.

Deliverables

- **One 17 x 11 inch landscape PDF sheet:** Lastname_Firstname_P2a.pdf. Include all numbered frames, adjacent instructions/pseudocode, exposed values, the final architectural view and source credits.
- **Editable graph:** .blend, or .gh plus .3dm where used. Internalize Grasshopper reference inputs; package any other necessary dependencies. A baked mesh or screenshots alone cannot demonstrate the operation history.
- Use Lastname_Firstname_P2a/ for the submission. Keep the object available for the separate P2b fabrication milestone.

Evaluation

- **CSG logic:** the required transforms and all three Boolean types produce the stated result and remain editable.
- **Reproducibility:** dimensions, axes, named operands and operation order make the procedure unambiguous.
- **Communication:** diagram and instructions agree; consistent views show one understandable change per frame.
- **Design and craft:** the spatial operation reads at building scale and the sheet follows the Graphic Standards Manual.

Before You Submit

- [] Eight or nine frames; at least three primitives; translation, rotation, scale, union, intersection and difference.
- [] Difference frames label the kept solid and cutter.
- [] Instructions and pseudocode match the graph and include parameter values.
- [] Final view includes a 1.75 m figure, vegetation, ground and shadow.
- [] One sheet and an editable source that opens with all dependencies.

P2b - CSG 3D printed object

Introduced: Week 3 - Sep 8 / **Due:** Week 10 - Oct 27 / **Weight:** 7%

Preparation checks: Week 6 - Sep 29: Check geometry, units, orientation and the sliced preview before queueing.

Description

Fabricate a variant of your P2a CSG object and explain the decisions that make it printable. Compare the final physical object with the digital geometry. Record any revision for closure, feature thickness or orientation so the fabrication copy remains traceable to the original construction.

Rules & Constraints

- Print a variant of your CSG object from P2a, within 60 mm on each axis and the 200 g allocation, without supports.
- Hand in the print in Week 10 with one process sheet showing orientation and a photograph.
- Submit the editable source, exported mesh and slicer project. Explain any geometry changes made for printability.

Parameters

- Uniform print scale and final X/Y/Z dimensions in millimeters. Each axis must be no greater than 60 mm in the selected print orientation.
- Printer/material profile, draft layer height, wall count and infill. Supports must be off. Record the chosen values rather than assuming a universal printer profile.
- Orientation, bed-contact face, estimated time and filament use. The total allocation is 200 g including failed attempts.

Required Studies

- **Geometry check.** Show the final fabrication mesh as a closed solid with consistent outward normals. Resolve open edges, internal faces and self-intersections before slicing.
- **Orientation comparison.** Compare two positions on the bed. Label contact areas, overhangs, bridges and unsupported islands; explain the chosen orientation. Revise geometry if no orientation is viable without supports.
- **Sliced and physical result.** Inspect the layer preview for missing thin features and floating starts. Record dimensions, settings, predicted time/material and a photograph of the completed object. Explain one difference between the prediction and the print.

Workflow

1. Duplicate/export the P2a result for fabrication while keeping its source graph intact. Record changes made to the geometry.
2. Export the mesh in millimeters and confirm all three dimensions in the slicer; STL itself does not store units. Scale uniformly to preserve proportions.
3. Complete the geometry/orientation/slicing check before entering the queue. Follow current FabLab orientation and machine procedures.
4. Print with supports off, keep a record of material used including failed attempts, and photograph the final object. Assemble the process sheet using the Graphic Standards Manual.

Deliverables

- **One physical CSG print** within 60 x 60 x 60 mm, produced without supports and within the 200 g allocation. Bring it for the Week 10 submission with your name on a separate label.
- **One 17 x 11 inch landscape process sheet:** Lastname_Firstname_P2b.pdf. Include the orientation comparison, dimensioned print view, slicer settings/time/material, geometry changes and a clear photograph of the completed print.
- **Editable source, exported mesh and slicer project** in Lastname_Firstname_P2b/. Include a screenshot showing dimensions and supports off. A preview or render does not replace the physical object.

Evaluation

- **Printability:** the submitted object and mesh satisfy the size, material and support constraints.
- **Fabrication reasoning:** orientation and any geometry changes follow a documented reading of the model and sliced layers.
- **Evidence:** the source, mesh, slicer project, process sheet and physical result correspond to the same variant.
- **Communication:** the sheet clearly records settings, observations and what you learned from the print.

Before You Submit

- [] Each dimension is at most 60 mm; supports are off; cumulative filament use is at most 200 g.
- [] Mesh closure, normals, thin features and layer preview have been checked.
- [] The process sheet includes orientation reasoning, settings and a photograph.
- [] Physical object, source, mesh and slicer project are ready for submission.

Shared standards: Course software, fabrication, attribution and AI policies apply. Submit through the channel announced in class/Canvas. Test the packaged source from a clean folder. Percentages are existing course weights. This handout uses US Letter; student sheets are 17 x 11 inches.